



A wind tunnel for anemometer calibration in the range of 0.2–1.25 m/s

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ABSTRACT

In order to calibrate different types of anemometers in a low-speed range (0.2–1.25 m/s) using pressure measurements, a wind tunnel has been built and characterized. The wind tunnel has a low-speed section and a high-speed section. The wind tunnel calibration has been performed by means of comparison of airspeed at the test cross section (low-speed) and at the reference cross section (high-speed). Measurements have been taken with a calibrated hot-wire anemometer and static and total probes respectively. A linear fitting is used to relate reference and test airspeed. A model based on mass conservation law has been developed. Calculations have been made with and without introducing the effect of the speed profile. Uncertainties of the experimental calibration are presented.

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1. Introduction

A wind tunnel for the calibration of anemometers in the **range of 0.2 to 1.25 m/s, using total and static pressure measurements as reference has been developed**. The intended application is the calibration of **hot-wire, hot-film and vane probe anemometers** used for measuring air flow in applications such as ambient air, air conditioning and ventilation ducts. A target uncertainty of 10% for airspeed calibration is considered acceptable.

The wind tunnel is an open circuit one; it comprises the reference and test sections, with different circular cross areas; one to the other joined by a contraction. The reference devices are placed in the high-speed test section, while the instrument to be calibrated is placed in the low-speed one.

Pitot tubes have been traditionally used as reference instruments for airspeed calibrations. Today, **Laser Doppler Anemometer (LDA) is also used as primary airspeed measurement system [1]**. The uncertainty of the LDA is good enough even at low airspeed but more expensive than other methods.

It is known that **measurements of airspeed below 2 m/s with Pitot tubes are not accurate**. The present device has been developed in order to perform measurements of airspeed in a simple way, i.e. by means of **static and total pressure as a reference instrument**.

This solution, a wind tunnel with a contraction, is not new. For instance, Omori [2] has developed a device for the calibration of

anemometers at low-speed air flow that consists of a wind tunnel and a shaped nozzle. The speed in the wind tunnel is increased by the nozzle and there measured. The anemometer is placed in the wind tunnel and the airspeed in that section is determined by conversion.

There are other methods to calibrate anemometers at low airspeed. Al-Garni [3] has developed a swinging arm that moves the probe in stagnant air. Bruun et al. [4] have studied that principle too.

Tsanis [5] and Chua et al. [6] mounted hot-wire probes on a traversing mechanism that is driven by a servo motor and travels at constant speed.

Yue and Malmström [7] developed a system that produces a simple laminar pipe flow generated by draining water from an air-tight container at a constant rate.

Cowdrey [8] described a method for calibrating low-speed hot-wire anemometers by using a small whirling arm. The instrument rotates when supported at the end of the arm, at a number of measured speeds in air initially at rest.

Sillanpää and Heinonen [9] developed a method based on the humidification, with a known mass of water, of flowing air before the blower on a wind tunnel. Then airspeed can be calculated as a function of humidification water flow.

Other researchers use flow meters as reference for the calibration at low speeds (Manca et al. [10], Seifert and Graichen [11]).

Ringegni et al. [12,13] have also analysed different solutions (traversing probe in still air, whirling arm, measurement of the displacement of a bubble in a wind tunnel) for the calibration of anemometers at low airspeed.

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Nomenclature

A [m^2]	Cross section area
K [dimensionless]	Blockage factor for a cylindrical stem
n [dimensionless]	Mean and centre airspeed ratio
Pd [Pascal]	Dynamic pressure
S [m^2]	Frontal projected area of the total tube stem
Q [m^3/s]	Volumetric flow rate
R [mm]	Duct radius
u [m/s]	Boundary layer airspeed
U [m/s]	Uncertainty
V [m/s]	Airspeed
$\bar{V}$ [m/s]	Average airspeed
x [mm]	Distance between the cross section and the inlet of the duct
y [mm]	Distance from the duct wall
δ [mm]	Boundary layer thickness
ν [m^2/s]	Kinematic viscosity

Subscripts

c	Centre of the section
r	Reference section
t	Test section
C	Combined
EXP	Expanded
FIT	Fit
REF	Reference airspeed
RES	Resolution
$RES-MICRO$	Micromanometer resolution
SNU	Section non-uniformity
∞	Border of the boundary layer

2. Experimental apparatus

2.1. Wind tunnel description

The wind tunnel has these parts, as shown in Fig. 1: an inlet contraction, a grid to attenuate flow fluctuations, a low-speed duct, a contraction, a high-speed duct and a blower.

The inlet contraction has a diameter ratio of 2.3:1. The low-speed duct is a 340 mm diameter PVC pipe. It has 1190 mm length. The test cross section is placed at a distance of 890 mm downstream the union with the contraction. To reduce speed fluctuations inside the tunnel, a flexible metal mesh is placed between the contraction and the low-speed duct. Then another contraction links the low and high-speed ducts. The high-speed duct tube is 1490 mm length and has a 100 mm diameter. The reference section is placed 9.5 duct diameters downstream the contraction. Four holes equally spaced on the surface of the duct allow to

perform measurements of static pressure. The total pressure tube, diameter 4 mm, is situated at the centre of the test cross section. A straightener is placed between the reference section and the centrifugal blower. Considerations stated by the AMCA Standard 210 [14] have been introduced in the design of the device. A photograph of the wind tunnel is shown in Fig. 2.

2.2. Instrumentation

The dynamic pressure has been determined measuring static and total pressure by means of an R-Fuess differential micro-manometer; range: 1600 Pa; resolution: 0.5 Pa. The air density has been determined measuring atmospheric pressure and dry and wet bulb temperatures. Atmospheric pressure has been measured with a Druck DPI-740 barometer; sensor type: resonant pressure transducer; accuracy: 0.02% FS; resolution: 1 Pa. Ambient temperatures have been measured by means of mercury liquid in glass thermometers; resolution: 1 °C.

The instrument used for the calibration of the wind tunnel is an Alnor AVM 430 hot-wire anemometer; calibrated, traceable to NIST [15,16]. Speed range: 0–30 m/s; resolution: 0.01 m/s; calibration uncertainty, see U_{CAL} in Table 4.

The flow rate is managed by a voltage regulator that controls the blower speed. Further information on instrumentation and wind tunnel details can be found in [17].

2.3. Method of calibration

The wind tunnel calibration has been carried out comparing the airspeed at the centre of the test cross section, measured with the hot-wire anemometer, and the airspeed at the reference cross section, obtained using total and static probes.

The procedure is as follows: the wind tunnel speed is set at a certain value with the voltage regulator, the flow stabilizes and then the measurements with total and static probes and hot-wire anemometer are taken. This procedure is repeated for different speeds within the wind tunnel range. The airspeed range at the test cross section varies from 0.18 to 1.25 m/s and the corresponding airspeed at the reference cross section from 2 to 14.2 m/s. Three repetitions have been done for each airspeed tested. The whole procedure has been repeated three different days under similar conditions.

Airspeed has been measured at the centre of the test cross section and at a distance of 40 mm from the centre in 3 points along radials separated 120°. A linear least square fitting for the values of all the tests performed has been done. The equation of this line is considered to be the wind tunnel calibration equation.

3. Theoretical approach

Although the wind tunnel has been calibrated comparing the total and static pressure measurements with those of a calibrated

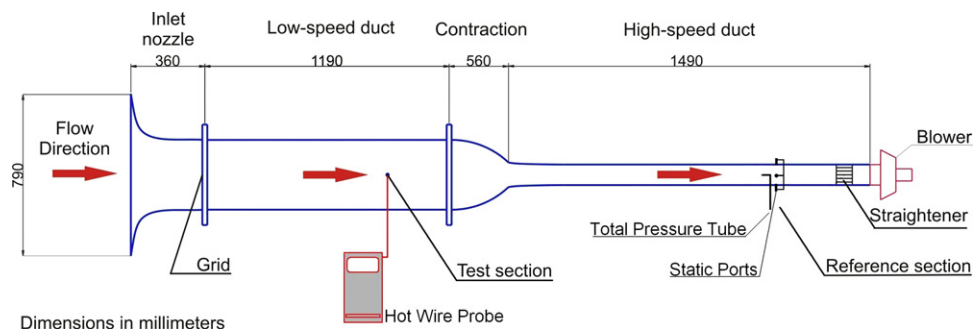


Fig. 1. Wind tunnel schematic diagram.



Fig. 2. Wind tunnel photograph.

hot-wire anemometer, a simple theoretical approach has also been developed. For this theoretical analysis, based on the mass flow conservation criteria, two different approaches have been used to relate the airspeed between the test and reference cross sections.

For the first case (Area ratio between ducts model), the mass flow conservation equation for constant density states: $Q = A_t \cdot \bar{V}_t = A_r \cdot \bar{V}_r$; where Q is the volumetric flow rate, A_t and A_r are the cross section areas for the test and reference sections respectively, and $\bar{V}_t$ and $\bar{V}_r$ are the average airspeed for the test and reference sections respectively. Then, the relation between the average airspeed for both cross sections is the area ratio between ducts: $\bar{V}_r = 11.56 \bar{V}_t$, which is the ratio between the airspeed at the centre of the test and reference sections.

The second theoretical case takes into account the airspeed profiles. The evolution of the flow along the ducts is affected by the boundary layer thickness, which has been calculated for the test and reference cross sections. Then the relation between airspeed at the centre of the test and reference sections is determined. Boundary layer thicknesses have been only calculated for lowest and highest airspeeds. It has been considered that boundary layers increase downstream from the inlet of each duct. According to that, boundary layers are laminar, except for the test cross section at the highest airspeed. The flow outside the boundary layers is turbulent for both conditions.

The boundary layer thicknesses have been calculated from [18]

$$\delta = 5.48 \sqrt{\frac{\nu x}{V_\infty}} \quad (1)$$

for the laminar boundary layer and

$$\delta = 0.37 \left(\frac{\nu}{V_\infty} \right)^{\frac{1}{5}} x^{\frac{4}{5}} \quad (2)$$

for the turbulent boundary layer, where δ is the boundary layer thickness, ν is the kinematic viscosity, x is the distance between the cross section and the inlet of the duct, and V_∞ is the airspeed at the border of the boundary layer.

The boundary layer airspeed profile is determined by

$$u = V_\infty \left[2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2 \right] \quad (3)$$

for the laminar boundary layer and

$$u = V_\infty \left(\frac{y}{\delta} \right)^{\frac{1}{7}} \quad (4)$$

for the turbulent boundary layer, where, u is the airspeed and y is the distance from the duct wall.

The cross section average airspeed is

$$\bar{V} = \frac{Q}{A} = \frac{2}{R^2} \int_0^R V_r r dr \quad (5)$$

where R is the duct radius and V_r is the airspeed at a distance r from the duct wall.

The relation between the average and the centre airspeed is given by $A_t \bar{V}_t = A_r \bar{V}_r = A_t n_t V_{ct} = A_r n_r V_{cr}$. Where V_{ct} and V_{cr} are the airspeed at the centre of the duct for the test and reference sections respectively and $n = \bar{V}/V_c$.

Then the ratios between the airspeed at the centre of the test and reference sections are: $V_{cr} = 11.83 V_{ct}$, for the lowest airspeed; and $V_{cr} = 12.25 V_{ct}$, for the highest airspeed.

4. Results and discussion

4.1. Wind tunnel calibration

Tables 1–3 show the wind tunnel calibration results for three tests performed in different days with the same procedure.

In order to determine the combined uncertainty of measurements, the standard deviation of the mean of each series of independent observations (A Type uncertainty), the calibration and resolution uncertainties of the hot-wire anemometer, U_{CAL} and U_{RES} respectively, the uncertainty of the reference airspeed determination, U_{REF} , the standard deviation of the adjustment line, U_{FIT} , and the airspeed profile non-uniformity uncertainty, U_{SNU} , have been taken into account.

U_{CAL} is the unexpanded uncertainty of the calibration of the hot-wire anemometer. It is given by the calibration certificate, emitted by HITEC Ingeniería [16].

U_{REF} includes the following uncertainty sources: the calibration and resolution of the micromanometer, the blockage effect of the total pressure tube and the measure of the tube areas.

The calibration uncertainty of the micromanometer is given by its calibration certificate.

The resolution uncertainty of the micromanometer is determined as

$$U_{RES-MICRO} = \text{Resolution} / \sqrt{12}, \quad (6)$$

standard deviation for a uniform probability distribution.

The blockage effect of the total pressure tube is calculated as a dynamic pressure addition given by $\Delta Pd = -0.7K(S/A)Pd$, where K is the blockage factor for a cylindrical stem, S is the frontal projected area of the portion of the stem immersed in the fluid and A is the cross sectional area of the duct.

The tube area uncertainty is determined from the measurement of three diameters along radials separated 120°.

The U_{REF} has been transferred from the high-speed tube to the low-speed tube by the area ratio.

The combined uncertainty has been calculated as follows [17]:

$$U_c = [(U_A)^2 + (U_{CAL})^2 + (U_{RES})^2 + (U_{REF})^2 + (U_{FIT})^2 + (U_{SNU})^2]^{1/2} \quad (7)$$

The values of A and B type uncertainties considered are shown in Tables 4 and 5.

The coverage factor adopted to determine the expanded uncertainty, U_{EXP} , is 2, confidence level of 95%, assuming that the combined uncertainty probability distribution is normal [19] (Fig. 3).

Table 6 shows the combined and expanded uncertainty for the tested airspeed range. The evolution of U_{EXP} is shown in Fig. 4.

The results of the three tests presented are shown in Fig. 4. Tests show good repeatability.

The least square fitting for the values of all the tests performed has been adopted as the wind tunnel calibration line, whose equation is $V_t = 0.085 V_r \pm 0.01 \text{ (m/s)}$.

Table 7 shows the difference between the experiments and the wind tunnel calibration line for the Day 1 tests. Similar differences have been found for the other tests.

Measurements performed at 40 mm from the duct centre show differences of 0.02 m/s or lower than the corresponding centre airspeed. The uncertainty is evaluated with U_{SNU} .

Table 1

Day 1 Test results.

Reference airspeed (m/s)	Test section airspeed			Mean (m/s)	Ucombined (m/s)	Uexpanded (m/s)
	1 (m/s)	2 (m/s)	3 (m/s)			
2.06	0.19	0.19	0.19	0.19	0.03	0.05
4.70	0.41	0.42	0.42	0.42	0.03	0.05
7.08	0.61	0.63	0.61	0.62	0.02	0.05
9.44	0.80	0.82	0.82	0.81	0.02	0.04
11.76	1.02	1.02	1.00	1.01	0.02	0.04
14.27	1.24	1.25	1.25	1.25	0.03	0.06

Barometric pressure: 101284 Pa; Dry bulb temperature: 20 °C.

Table 2

Day 2 Test results.

Reference airspeed (m/s)	Test section airspeed			Mean (m/s)	Ucombined (m/s)	Uexpanded (m/s)
	1 (m/s)	2 (m/s)	3 (m/s)			
2.26	0.20	0.20	0.20	0.20	0.03	0.05
4.70	0.42	0.42	0.42	0.42	0.03	0.05
7.32	0.64	0.64	0.63	0.64	0.02	0.05
9.36	0.81	0.82	0.81	0.81	0.02	0.04
11.71	1.02	1.02	0.99	1.01	0.02	0.05
14.29	1.25	1.24	1.23	1.24	0.03	0.06

Barometric pressure: 101334 Pa; Dry bulb temperature: 21 °C.

Table 3

Day 3 Test results.

Reference airspeed (m/s)	Test section airspeed			Mean (m/s)	Ucombined (m/s)	Uexpanded (m/s)
	1 (m/s)	2 (m/s)	3 (m/s)			
2.17	0.20	0.20	0.20	0.20	0.03	0.05
4.76	0.42	0.42	0.42	0.42	0.03	0.05
7.16	0.63	0.62	0.63	0.63	0.02	0.05
9.47	0.81	0.82	0.82	0.82	0.02	0.04
11.69	1.00	1.02	1.00	1.01	0.02	0.04
14.55	1.25	1.27	1.27	1.26	0.03	0.06

Barometric pressure: 101245 Pa; Dry bulb temperature: 21.5 °C.

Table 4

A type uncertainties.

Airspeed (m/s)	Day 1 test U_A (m/s)	Day 2 test U_A (m/s)	Day 3 test U_A (m/s)
0.2	0.0000	0.0000	0.0000
0.4	0.0027	0.0000	0.0000
0.6	0.0054	0.0027	0.0027
0.8	0.0054	0.0027	0.0027
1.0	0.0054	0.0082	0.0054
1.2	0.0027	0.0047	0.0054

Table 6

Combined and expanded uncertainties.

Airspeed (m/s)	U_C (m/s)	U_{EXP} (m/s)
0.2	0.013	0.026
0.4	0.014	0.029
0.6	0.021	0.043
0.8	0.024	0.048
1.0	0.025	0.050
1.2	0.028	0.056

4.2. Theory against experiments

In Table 8 the comparison between the experiments and the theoretical relation given by the area ratio between ducts (first theoretical approach) for the Day 1 Test is presented.

In Table 9, the comparison between experiments and theory is shown for the second theoretical approach. Here the theoretical ratio between the airspeed at the centre of the test and reference sections, taking into account the duct airspeed profiles, have been calculated for the lowest and the highest airspeeds.

Table 7

Difference between the experiments and the calibration equation for the Day 1 Test results.

Reference airspeed (m/s)	Mean airspeed (m/s)	Calibration equation (m/s)	Difference (%)
2.06	0.19	0.19	2.6
4.70	0.42	0.41	1.8
7.08	0.62	0.61	0.8
9.44	0.81	0.81	0.1
11.76	1.01	1.01	0.4
14.27	1.25	1.22	1.9

Table 5

B type uncertainties.

Airspeed (m/s)	U_{CAL} (m/s)	U_{RES} (m/s)	U_{REF} (m/s)	U_{FIT} (m/s)	U_{SNU} (m/s)
0.2	0.002	0.0029	0.006	0.0101	0.0040
0.4	0.006	0.0029	0.007	0.0101	0.0033
0.6	0.016	0.0029	0.009	0.0101	0.0030
0.8	0.016	0.0029	0.012	0.0101	0.0074
1.0	0.015	0.0029	0.015	0.0101	0.0064
1.2	0.018	0.0029	0.017	0.0101	0.0075

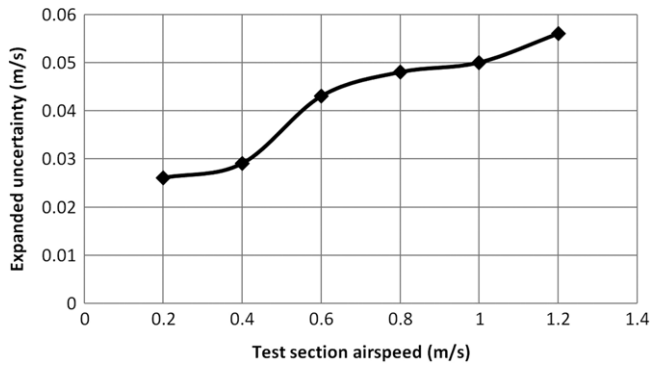


Fig. 3. Expanded uncertainty evolution.

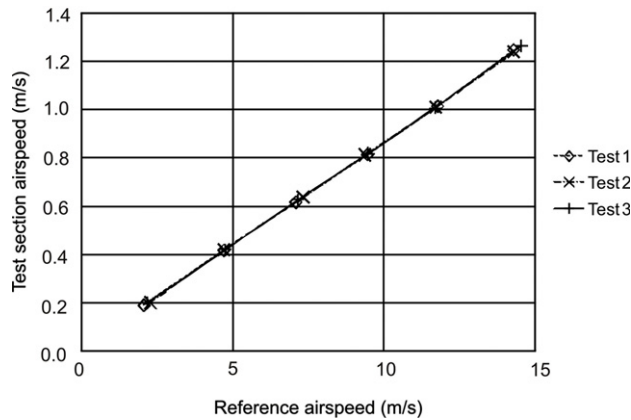


Fig. 4. Test results.

Table 8

Comparison between the experiments and the theoretical relation given by the area ratio between ducts (first theoretical approach).

Reference airspeed (m/s)	Mean airspeed at the test section (m/s)	Theoretical calculation (m/s)	Difference (%)
2.06	0.19	0.18	6.6
4.70	0.42	0.41	2.5
7.08	0.62	0.61	0.7
9.44	0.81	0.82	−0.4
11.76	1.01	1.02	−0.4
14.27	1.25	1.23	1.0

Table 9

Comparison between experiments and theoretical relation developed from the airspeed profiles.

Reference airspeed (m/s)	Average airspeed at the test section (m/s)	Theoretical calculation (m/s)	Difference (%)
2.06	0.19	0.18	5.5
14.27	1.25	1.19	4.8

This theoretical approach gives greater differences at the highest airspeed than the first model.

5. Conclusions

A simple and inexpensive wind tunnel for anemometer calibration in the range of 0.2–1.25 m/s has been built and calibrated. The highest expanded uncertainty obtained from the calibration in the range of operation is 0.06 m/s. The goal to attain an expanded uncertainty less than 10% for the wind tunnel calibration has been fulfilled. The device has been verified to be a suitable instrument for the calibration of anemometers in the specified range.

The theoretical model shows differences with the experimental measurements. The origin of these differences could be due to the simplicity of the model, which does not represent exactly the actual wind tunnel. These values are within expanded measurement uncertainties. The first theoretical approach (area ratio between ducts) is consistent with the experimental calibration except at low airspeeds. The second theoretical approach (including airspeed profiles) presents errors of approximately 5%.

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